Q.1 Two surfaces are said to be orthogonal at the point of their intersection if their normals at that point are ---------

A. in opposite direction
B. Overlapping
C. Parallel
D. Perpendicular

Q.2 If two dogs are pulling a bone with force=20Newtons in opposite direction, then the resultant force is---------

A. -40 Newton
B. 40 Newton
C. Zero Newton
D. -20 Newton

Q.3 Two vectors cannot have the same magnitude.

A. False
B. True

Q.4 Two vectors A and B are _____ if one vector A is scalar multiple of the other vector B.

A. parallel
B. opposite
C. perpendicular
D. equal

Q.5 Gradient of a scalar function always results in a.....function.

A. discontinuous
B. continuous
C. vector
D. scalar

Q.6 If f has a local maximum or minimum at c, then c is a _____ of f

A. critical number
B. maximum number
C. None of these
D. minimum number

Q.7 Given a function f(x,y), what is the definition of normal line to the surface z=f(x,y)?

A. A line parallel to the x-axis
B. A line perpendicular to the tangent plane at a point on the surface
C. A line that lies entirely in the tangent plane
D. A line parallel to the y-axis

Q.8 Which of the following is the slope if the equation of a line is given as □=−2/3□+5y?

A. -5
B. 5
C. 2/3
D. -2/3

Q.9 A zero vector has magnitude equal to zero and ---------- direction.

A. no
B. downward
C. upward
D. arbitrary

Q.10 {\text{If}}\,f(x,\,y)\,\,{\text{has}}\,{\text{a}}\,{\text{relative extrema at}}\,(x_0 ,\,y_0){\text{, then}}\,

A. f_x (x_0 ,\,y_0)\, < \,0,\,f_y (x_0 ,\,y_0)\, < \,0
B. f_x (x_0 ,\,y_0)\, < \,0,\,f_y (x_0 ,\,y_0)\, > \,0
C. f_x (x_0 ,\,y_0)\, > \,0,\,\,f_y (x_0 ,\,y_0)\, > \,0
D. f_x (x_0 ,\,y_0)\, = \,0,\,f_y (x_0 ,\,y_0)\, = \,0
